

Venkatesh Guttikonda

Artificial Intelligence and Machine Learning Engineer

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SUMMARY

Experienced Data Engineer and Data Analyst with over four years of expertise in developing robust data pipelines, predictive analytics, and machine learning models, now transitioning into AI/ML Engineering. Proficient in Python, SQL, and cloud platforms (AWS, GCP, Azure), with a strong background in big data technologies like Apache Spark and Kafka. Skilled in leveraging PyTorch to drive impactful results in both data analytics and emerging AI applications. Demonstrates a unique ability to apply data engineering principles to AI/ML challenges, bridging the gap between traditional data roles and cutting-edge artificial intelligence solutions. Proven track record of collaborating with cross-functional teams to deploy effective data and AI systems, enhancing overall business intelligence and operational efficiency.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL
- **Frontend Technologies:** ReactJs, Streamlit, NextJS
- **Data Manipulation and Analysis:** Pandas, NumPy, Scikit-Learn,
- **Data Visualization Tools:** Matplotlib, Seaborn, Tableau, Power BI, Excel ,Plotly
- **Big Data Technologies:** Apache Spark, Kafka, PySpark ,Snowflake, Apache Airflow
- **Machine Learning Frameworks and Libraries:** Langchain, NLP, PyTorch, RAG, HF Transformers
- **Cloud Computing:** AWS, Google Cloud Platform, Azure , Google Vertex AI
- **DevOps:** Docker, Kubernetes, CI/CD Pipelines, Jenkins
- **Databases:** NoSQL, MongoDB, Cassandra, Postgres,
- **Vector Databases:** Pinecone, Qdrant, Chroma, Weaviate, FAISS
- **Development Tools and Environments:** Jupyter Notebook, PyCharm, VS Code
- **Chatbot Framework:** Google Dialogflow, IBM Watson
- **Deep Learning :** CNNs, RNNs, LSTMs, and Transformers
- **Natural Language Processing:** Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Semantic Search, Entity Linking, Question Answering

PROFESSIONAL EXPERIENCE

X Node Inc.

Remote

AI/NLP Engineer

Aug 2024 – Present

- Train and fine-tune large language models using both structured and unstructured datasets to drive domain-specific conversational tasks.
- Design and develop robust data pipelines for agentic systems, including integrating human-in-the-loop feedback loops to continuously improve system performance.
- Architect scalable data infrastructure by integrating Azure Blob Storage, Azure Data Lakes, and modern databases to support complex AI agent interactions.
- Built hybrid retrieval-augmented generation (RAG) systems using LangChain and Azure AI, resulting in a 30% improvement in retrieval accuracy and response time for customer-facing NLP applications.
- Applied graph-based data structures and Neo4j to enhance AI models' decision-making capabilities, enabling smarter, context-aware AI systems for dynamic, real-time applications.
- Integrated Azure Machine Learning and Azure OpenAI services to further enhance AI capabilities, including Speech to text and Text to Speech(TTS).
- Optimized the use of vector databases (e.g., Qdrant, Pinecone) to efficiently store and retrieve embeddings, ensuring high-performance search and retrieval.
- Developed and deployed ReAct Agents using LangGraph to handle complex, multi-step reasoning tasks, improving decision-making workflows.

- Streamlined CI/CD pipelines for deploying containerized NLP models on Azure Kubernetes Service, reducing deployment time.
- Collaborated with data scientists, software developers, and stakeholders to ensure successful AI implementation.

Client: Walmart
Contract
Bentonville, Arkansas
Data Engineer

June 2023 – July 2024

- Collaborated with cross-functional teams to design and implement data solutions that met critical business needs, resulting in a 30% improvement in data-driven decision-making processes.
- Developed and maintained robust data pipelines and ETL processes using PySpark and Scala, processing over 100GB of data daily with high reliability.
- Designed and optimized data models and architecture, reducing data storage costs by 25% while boosting query performance.
- Implemented data integration and transformation workflows that improved data quality and ensured consistency across multiple sources.
- Monitored and troubleshooted data pipelines to maintain system uptime and expedite issue resolution.
- Worked closely with data scientists and analysts to provide tailored datasets and analytical tools, accelerating insights generation by 60%.
- Deployed a real-time data streaming solution using Spark Streaming and GCP Pub/Sub, enabling near real-time analytics.

Capital Numbers

Noida, India
Data Analyst

June 2019 – May 2021

- Engineered a Python-based predictive model to enhance product recommendations, boosting customer engagement through sophisticated machine learning algorithms and data analysis techniques.
- Developed custom data pipelines using Apache Spark and Kafka to ingest and process large-scale structured and unstructured data.
- Engineered custom ETL/ELT pipelines using Apache Spark and Kafka, consolidating large-scale data from diverse sources with high reliability.
- Conducted comprehensive statistical analyses to identify trends, optimize data formats, and inform predictive modeling for data-driven decision-making.
- Explored graph database concepts and spatial data analysis techniques, laying the groundwork for future GIS integration to create map markers and correlate geospatial data.
- Applied data preprocessing and feature engineering techniques, reducing data preparation time by 20% with Pandas and NumPy.
- Collaborated with cross-functional teams to define and prioritize data-driven business solutions, deploying models with TensorFlow and improving application efficiency.
- Conducted exploratory data analysis and visualized data trends, resulting in a 25% increase in actionable insights for strategic decision-making using Matplotlib and Seaborn.
- Automated model evaluation and performance tracking processes, reducing manual oversight with custom Python scripts.
- Engaged in code reviews and best practice development, contributing to a 15% increase in project efficiency and code quality.

PROJECTS

AI-Powered Document Understanding & Summarization System

- Engineered an advanced document processing pipeline using Hugging Face models and Azure OpenAI for automated document summarization.
- Integrated retrieval-augmented generation (RAG) with Weaviate vector database, achieving 50% faster retrieval of relevant context for summarization.
- Applied transformer-based extractive and abstractive summarization techniques, reducing manual summarization workload by 40%.

Conversational AI with Multi-Step Reasoning & ReAct Agents

- Designed and deployed ReAct Agents using LangGraph, enabling intelligent multi-step reasoning for enterprise workflows.
- Built a custom Agent Playground, allowing users to create and configure AI-driven automation workflows for industry-specific use cases.

AI-Driven Natural Language to SQL Query Engine

- Built an interactive Streamlit-based web app enabling natural language to SQL conversion, powered by Google Generative AI and LangChain.
- Designed an adaptive SQL query refinement mechanism using memory-enhanced prompt engineering, reducing query errors.
- Improved multi-turn conversational query execution, making database interactions more intuitive for non-technical users.

EDUCATION

Master of Science in Applied Computer Science

[Northwest Missouri State University](#)

August 2021 – May 2023

Bachelor of Engineering in Electronics and Communications

[Andhra University](#)

July 2015 – May 2019

